

PATIENT GUIDE

Keratoconus

Understanding, managing and treating:
A complete guide for patients of Samyak Drishti Eye Clinic.



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This guide is for patient education only and does not replace personalised medical advice. Consult your doctor at Samyak Drishti Eye Clinic for guidance specific to your eyes and health.

WHAT IS KERATOCONUS?

Keratoconus is a progressive eye condition in which the cornea, the clear, dome-shaped front surface of your eye, gradually thins and bulges outward into a cone-like shape. This distorts the way light enters the eye, causing blurred and distorted vision.

It typically begins in the teenage years or early twenties and may progress for 10–20 years before stabilising naturally. It can affect one or both eyes, though it is usually more severe in one eye.

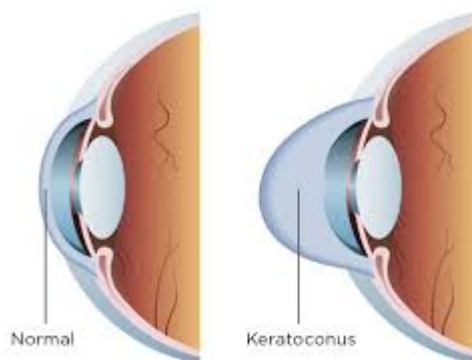


Diagram: Normal vs cone-shaped cornea

How the Normal Cornea Works

A healthy cornea is round and smooth, acting like a camera lens to focus light precisely onto the retina at the back of your eye. The precise curvature of the cornea is maintained by tiny protein fibres called collagen. In keratoconus, these collagen fibres weaken, reducing the cornea's ability to hold its shape.

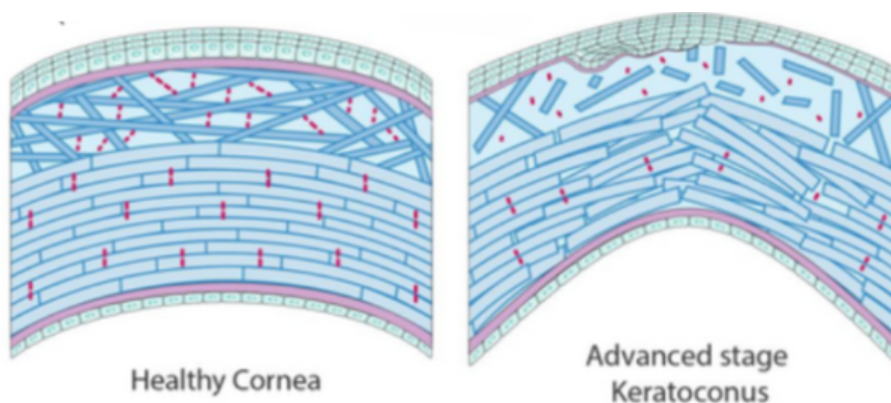


Diagram: Collagen fibre arrangement

Key Fact

Keratoconus affects approximately 1 in 2,000 people worldwide. It is not a rare condition, and it is very treatable, especially when diagnosed early.

CAUSES AND FACTORS

The exact cause of keratoconus is not fully understood, but research points to a combination of genetic, environmental, and biochemical factors.

Who Is at Greater Risk?

- Family history of keratoconus (first-degree relatives have a significantly higher risk)
- Chronic eye rubbing a major modifiable risk factor; rubbing releases enzymes that weaken corneal collagen
- Atopic conditions such as hay fever, asthma, and eczema
- Down syndrome (trisomy 21)
- Connective tissue disorders (e.g., Marfan syndrome, Ehlers-Danlos syndrome)
- Poorly fitting contact lenses worn over a long period

Important: Stop Rubbing Your Eyes

If you or a family member has been diagnosed with keratoconus, avoiding eye rubbing is the single most important lifestyle change you can make. Eye rubbing accelerates corneal thinning and can significantly worsen the condition.

Treat the underlying allergy with antihistamine eye drops or oral medication to reduce the urge to rub.



Keratoconus often starts subtly. Early symptoms can be mistaken for ordinary short-sightedness or astigmatism. It is important to see an eye specialist promptly if you notice any of the following:

Common Symptoms

- Blurred or distorted vision that is not fully correctable with glasses
 - Increasing short-sightedness (myopia) or astigmatism a change in your glasses prescription every few months
 - Sensitivity to bright light and glare (photophobia)
 - Seeing halos around lights, especially at night
 - Difficulty driving at night
 - One eye appears noticeably worse than the other
 - Ghost images or 'starbursts' around lights
-



Diagram: Common Symptoms are Halos and Ghost images

When to See a Doctor Urgently

Sudden clouding of vision, severe eye pain, or rapidly worsening sight may indicate acute hydrops, a rare complication where the cornea absorbs fluid and swells suddenly. This requires same-day medical attention.

DIAGNOSIS

Keratoconus cannot be diagnosed by a standard sight test alone. Your specialist at Samyak Drishti will perform a comprehensive assessment including:

Diagnostic Tests

- Corneal Topography: a detailed map of the cornea's surface curvature, the cornerstone of diagnosis
- Corneal Tomography (Scheimpflug imaging): 3D mapping of both the front and back surfaces, revealing thinning not visible on topography alone
- Corneal Biomechanics Testing: One of the most advanced steps in modern keratoconus evaluation. Using a gentle air puff, devices like the Corvis ST measure how the cornea responds to pressure, with parameters such as corneal stiffness and deformation helping to identify keratoconus at its earliest stage.
- Pachymetry: measurement of corneal thickness at multiple points
- Best-corrected visual acuity: how well you see with optimal glasses
- Slit-lamp examination to detect Vogt's striae (stress lines in the cornea), Fleischer ring (iron deposits), and corneal scarring

When corneal tomography and biomechanics testing are used together, they provide a far more complete picture than either test alone, catching what one would miss without the other.

Why Us?

With corneal topography, Pentacam tomography, and Corvis biomechanics all under one roof, Samyak Drishti offers one of the most comprehensive keratoconus

diagnostic suites in Mumbai. Combined with Dr. Sujal's expertise in interpreting these tests, you can be confident that no early sign goes undetected.

Staging Keratoconus

Keratoconus is graded from Stage 1 (mild, early) to Stage 4 (advanced, with significant thinning and scarring). Knowing the stage guides treatment. Early detection is key to preserving vision.

TREATMENT OPTIONS

The goals of treatment are twofold:

- 1. Stop the progression of keratoconus
- 2. Improve vision quality

I. To Improve Vision Quality

Treatment	Purpose	Best For
Spectacles / Soft Contact Lenses	Correct mild refractive error in early disease	Early/mild KC
Rigid Gas Permeable (RGP) Lenses	Vault over the irregular cornea, improving optics	Mild to moderate KC
Scleral Lenses	Large lenses resting on the sclera for better comfort and vision	Moderate to advanced KC
Topography-Guided PRK + CXL	Regularise corneal shape and stabilise	Early to moderate KC
Implantable Collamer Lens (ICL)	Correct residual refractive error after stabilisation	Post-CXL stable KC
Corneal Transplant (DALK / PKP)	Replace diseased corneal tissue	Advanced / scarred KC

II. To Improve Stability

Treatment	Purpose	Best For
Corneal Collagen Cross-Linking (CXL)	Halt progression by stiffening corneal fibres	Progressive KC at any stage

CORNEAL COLLAGEN CROSS-LINKING

Cross-linking is the most significant advance in keratoconus management in decades. It is the only treatment proven to halt disease progression early. At Samyak Drishti, we perform epithelial-off CXL, the gold-standard, clinically validated technique.

How CXL Works

Your cornea naturally contains collagen fibres held together by chemical cross-links that give it strength and shape. In keratoconus, these links are deficient. During CXL, riboflavin (vitamin B2) drops are applied to the corneal surface and activated with a precise dose of ultraviolet-A (UVA) light. This triggers a photochemical reaction that forms new covalent bonds between collagen fibres, substantially stiffening the cornea.



What to Expect: Step by Step

- The surface epithelium is gently removed (epithelial-off technique) to allow riboflavin penetration
- Riboflavin drops are instilled every 2–3 minutes for approximately 30 minutes
- The cornea is then exposed to calibrated UVA light for 30 minutes
- A protective contact lens bandage is placed on the eye
- Total in-clinic time is approximately 90 minutes

After CXL: Recovery Timeline

Days 1–3: Discomfort, light sensitivity, and blurred vision while the surface heals. Prescribed drops and pain relief are provided.

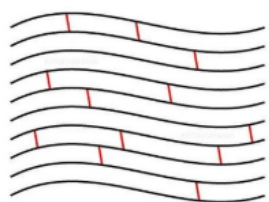
Week 1: Vision remains variable; most patients return to light activities.

Months 1–3: Vision may fluctuate. This is normal as the cornea is stabilising.

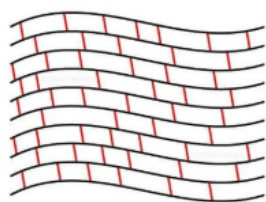
Months 3–12: Vision gradually stabilises. Topography is repeated to confirm halted progression.

Success rate: CXL halts progression in over 95% of patients. In many, some improvement in corneal shape is seen.

Before CXL: less crosslinking



After CXL: more crosslinking

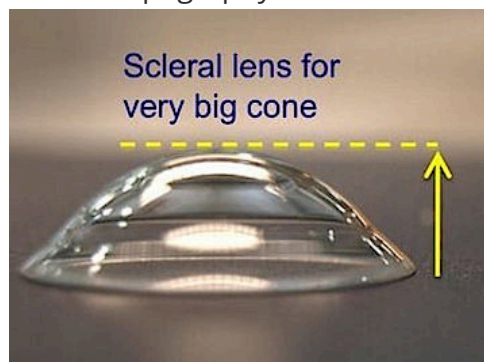
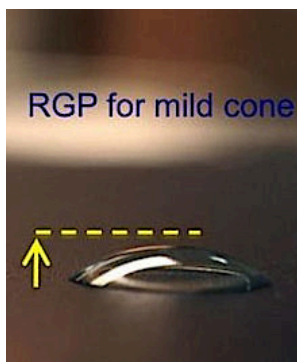


Living with Keratoconus

Keratoconus is a lifelong condition, but with the right management, most people lead full and active lives with good vision. Here is how to work with your condition day to day.

Contact Lenses & Vision Correction

For many patients, specialist contact lenses remain the best way to achieve clear vision, particularly rigid gas permeable (RGP) lenses (for early stages) and scleral lenses (for late stages). These vault over the irregular corneal surface, creating a smooth optical interface. Your optometrist will custom-fit lenses to your corneal topography.



Protecting Your Eyes

- Use antihistamine eye drops if you have allergies to reduce the urge to rub
- Wear UV-protective sunglasses outdoors
- Attend all follow-up appointments even if vision feels stable to diagnose early progression

Monitoring & Follow-Up

After CXL or any intervention, repeat corneal topography/tomography is performed at 3, 6, and 12 months, then annually. The aim is to confirm stability. If progression is detected, further intervention is discussed.

Disclaimer

This guide is for patient education only and does not constitute medical advice. Diagnosis and treatment decisions must be made by a qualified ophthalmologist after a thorough clinical assessment. Please consult your Samyak Drishti specialist for personalised guidance.

CONTACT & APPOINTMENTS

Book Your Keratoconus Consultation

Don't let discomfort become your normal.

At Samyak Drishti, you will receive a thorough tear film evaluation, a clear diagnosis, and a personalised treatment plan suited to your lifestyle and needs.

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